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**The Road to Hyperinflation: Economic
Independence in Ukraine, 1991-1993¹**

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ABSTRACT

The economic performance of independent Ukraine has been disappointing.

While being adversely affected by disruption of trade and higher energy prices related to the breakup of the Soviet Union, Ukraine has been less badly affected than most other ex-republics. Nevertheless, Ukraine has one of the worst inflation rates in the region and in the second week of August 1993 the black market exchange rate of the Ukrainian coupon was depreciating up to 7 percent a day against the dollar.

The main reason for Ukraine's poor performance has been its government's macroeconomic policies. Credit provided at negative real interest rates and a growing government budget deficit have meant high and rising rates of monetary emission. De facto price controls continue to distort markets and most trade in food products remains in the state sector. There has also been relatively little progress with privatization.

The failure to control inflation has made it more difficult to establish a strong system of public finance, to start meaningful private sector development and to restructure the banking system. Ukrainian experience demonstrates what happens when partial and gradual anti-inflation measures fail. In summer 1993 it appears that only a tougher and more radical set of policies can avert hyperinflation, but no political leader seems willing to adopt these measures.

I. INTRODUCTION

When independence was declared in August 1991 and confirmed by a referendum that December, Ukraine was initially considered by both Ukrainians and outsiders to have relatively bright economic prospects. Ukraine's well educated workforce, large size, endowment of natural resources and high level of industrialization appeared to confer important advantages relative to other post-communist countries. Western Ukraine had experienced a market economy more recently than most of the former Soviet Union and there was talk of Ukraine doing better than Russia and perhaps even joining the "fast-track" of economic reform seen in Poland, Hungary and the Czech and Slovak Federated Republic.

Two years later the situation is quite different. Ukraine has one of the highest inflation rates in the region and appears to have moved into a phase of hyperinflation. At the same time, the fall in output has been severe, there has been very little progress with privatization and the state remains heavily involved in the economy. Key policymakers give the impression they would prefer a return to the planned system, but under difficult circumstances they have settled for a very poorly functioning hybrid of market and intervention. Ukraine is now regarded almost universally as a disappointing case of failed reform in the former Soviet Union.²

Some might argue this is an unfair assessment. Certainly Ukraine has suffered both a large dislocation to its trade with other former republics and a negative terms of trade shock due to the higher price of imported energy. But most other newly independent countries formerly in the Soviet Union were either more involved in inter-republican trade or more dependent on energy imported from Russia. Compared to countries with similar or worse structural economic problems, Ukraine stands out as a bad performer. Why and what really went wrong in Ukraine?

The answer to both questions lies with inflation. Inflation was a significant and growing problem in 1991 for all parts of the Soviet Union. Ukrainian inflation during 1992 was over 1,000 percent and roughly in line with that in Russia — not surprising since Ukraine remained at least partially in the ruble zone. However, inflation has accelerated dramatically since Ukraine left the ruble zone fully in November 1992. In the 7 months up to November 1992, prices in Ukraine increased by 135 percent. In the 7 months since November, prices have risen 1,863 percent. From February to May 1993, monthly inflation was in the range of 15-23 percent. In June a further acceleration

² The recent coal miners strike in the Donbass region is the leading example of Ukrainians' discontent. The scathing review of the Ukrainian economy by PlanEcon is but one case of a negative western assessment of economic performance.

of inflation pushed the monthly rate over 100 percent, although this fell to 21 percent in July.

Data collected by us on black market exchange rates reveals a matching depreciation. The Ukrainian coupon is one of the few currencies which has fallen consistently against the ruble over the past 6 months: from 1.5 in November 1992 to 4.4 on August 4. In the month to August 11, the coupon fell 83 percent against the dollar and in the second week of August it was depreciating by up to 7 percent a day (see Table 1). Ukrainian is slipping into hyperinflation.

There has also been a fall in output in almost all sectors. Real GDP in the first quarter of 1992 was about 20 percent below the real GDP of the first quarter of 1991. Real GDP in the first quarter of 1993 showed a further fall of 10 percent compared to the same period in the previous year (Popiel 1993b, p. 19). Such measures are not completely reliable when inflation is so high, but the dismal picture is confirmed by physical measures of sectoral output.

Measured real wages have also fallen and there has been a particularly large decline in purchasing power over basic goods, such as bread and meat. Our own surveys of state stores have not found significant improvements in the availability and quality of food. (Details on these surveys are contained in Johnson and Ustenko 1992 and 1993d.)

Why has Ukrainian economic performance been so bad? Certainly there were adverse initial shocks over which Ukraine had no control, particularly the disruption of inter-republican trade and the increase in energy prices.

However, in large part blame must be attributed to lack of government strategy for economic adjustment. There have been piecemeal and gradual reforms but the evidence indicates these are far from being enough.

As a result, Ukraine has moved rapidly from a situation in which inflation is a result of underlying difficulties to a situation in which it is so high and unstable that it threatens to become the major problem.

Independent Ukraine's path towards hyperinflation is the primary subject of this paper. **Section 2** provides background economic statistics on the Ukrainian economy and relevant comparisons with other former republics. **Section 3** traces the major developments of the period of partial monetary independence, from December 1991 to November 1992. **Section 4** analyzes the implications of full monetary independence after November 1992. **Section 5** discusses the reasons for the disappointing pace of privatization in Ukraine. **Section 6** assesses the extent to which structural changes are taking place in the economy. **Section 7** assesses the impact of independence on real wages and living standards while **Section 8** concludes with the likely prospects for Ukraine.

II. THE STARTING POINT³

Ukraine definitely has economic potential. In contrast to most newly independent countries in the region Ukraine is quite large. Its population is estimated to be 51.8m people, which is about 18 percent of the former Soviet Union population (IMF Table 3). Its geographical area is 603.7 thousand square kilometers, slightly smaller than Texas or Turkey but larger than France. Furthermore most Ukraine's land is well endowed with either mineral resources or good quality soil. These facts are reflected in Ukraine's relatively large share in industrial and agricultural output of the former Union.

In 1989, Ukraine had 24 percent of Union coal production (IMF 1991, Table 11), 46 percent of iron ore production (IMF 1991, Table 12), 16.2 percent of total net output (IMF 1991, Table 13) and 17.2 percent of agricultural net output (IMF 1991, Table 13). It had more than 30 percent of Union production of the following industrial products (IMF 1991, Table 16): cast iron (40.8 percent), steel (34.2 percent), rolled ferrous metal (34.5 percent), steel pipe (33.5 percent), A.C. electric motors (35.8 percent), televisions (35.9 percent), sugar (52.6 percent), vegetable oil (33.2 percent). It also had between 20 percent and 30 percent of all Union production in a variety of machine building and equipment products, some construction materials, shoes and meat products. Furthermore, Ukraine also produced more than 20 percent of Union agricultural production in almost all major agricultural products, for example: grain (24.3 percent), potatoes (24.5 percent), vegetables (26.2 percent), meat (22.4 percent), milk (22.7 percent) and sugar beets (54.2 percent).

Ukraine did not conduct a large amount of trade with countries outside of the Soviet Union before independence. In 1990, its foreign exports, valued at international prices were 7,829m rubles, which was about 5 percent of GDP. To put this in international perspective, France exported 23 percent of its GDP in 1989, up from 13 percent in 1965, while Turkey — which has a population of around 55m and a land area of 779 square kilometers — exported 6 percent of its GDP in 1965 and 22 percent in 1989.

Imports from abroad into Ukraine in 1990 were 9301m rubles, implying a significant deficit. However, western agencies estimate that the switch to international prices actually improves Ukraine's trade balance with the rest of the world. Unfortunately, moving to world prices for trade with other former republics significantly worsens Ukraine's trade balance.

3 A detailed study of the Ukrainian economy near the end of the communist period is provided by Koropecjy 1992. Ukraine at the time of independence is assessed by U.S. Government 1992 and Hogan 1991.

Using Soviet domestic prices, Ukraine had a small inter-republican trade surplus in 1990. Its exports to other republics were 38,319 million rubles, which was about 23 percent of GDP, and its imports were 38,989 million rubles. Evaluated at world prices, those same exports would cost closer to 35,000 million rubles and the imports would cost over 42,000 million rubles. Oil and gas combined were a substantial deficit item but were partly made up for by net exports from the ferrous metals sector, the machine building sector and food processing.

Most other former republics exported a far higher proportion of their exports to other parts of the Soviet Union. Expressed as a proportion of net material product in 1988, Ukrainian inter-republican exports were 39.1 percent. In contrast, the Baltics exported 60-67 percent, Armenia exported 63.7 percent, Belorussia exported 69.6 percent, and Central Asia countries exported 41-51 percent. Only Kazakhstan and Russia were relatively less involved in inter-republican trade — they exported 30.9 percent and 18 percent respectively to other republics.

One estimate puts the total terms of trade effect on Ukraine of the move to world prices at a loss of around 2.5 percent of GDP. However, that estimate does not take into consideration the costs of disrupting inter-republican trade. In the first 9 months of 1992, inter-republican exports were around 575 bn rubles, which was about 20 percent of GDP, while foreign exports were still around 5 percent of GDP. Ukraine may therefore have lost as much as a fifth of its exports, primarily through difficulties with exports to Russia.

The most important trade issue is how much Ukraine will pay for oil and natural gas from other former republics, and how much it will charge for the use of the gas and oil pipelines which cross its territory on the way to more Western parts of Europe. For a former Soviet state Ukraine is unusually self-sufficient in energy, due to both its nuclear power stations and its coal. In 1990, about half of Ukraine's primary energy consumption was provided by its own resources, although this proportion had fallen from around 60 percent 5 years previously. In 1991, Ukraine imported about half of its energy needs primarily from Russia.

Ukraine's most important energy problem is its almost complete reliance on imported oil, although it has its own large processing capability. There have been shortages of gasoline and aviation fuel sporadically since independence and chronically since the end of 1992. Ukraine is currently paying 40-70 percent of world prices for various energy sources but there is an agreement in principle to pay world prices for Russian oil from the beginning of 1994. However, in early summer 1993 Ukraine was falling behind in its payments for Russian oil and as a result deliveries were down to one-fifth of the agreed level (Transition, p. 13).

No account of Ukraine's initial starting position would be complete without mentioning nuclear weapons. Inheriting the third largest nuclear stockpile in the world

has certainly helped Ukraine grab some headlines and attain some strategic importance — see, for example, Mearsheimer 1993, Economist 1993a and 1993b, The New York Times 1993. Unfortunately, Ukraine's position on these weapons has so far delayed rather than speeded up the building of links with the west, particularly in terms of obtaining western economic assistance. However, as the subsequent sections make clear, the confused macroeconomic situation in Ukraine itself makes it difficult for western agencies to help in any meaningful way.

Ukraine may have considerable economic potential, but its performance to date has been poor. The government's GDP forecast in April 1993 was 13,658 billion karbovantsi for calendar 1993. The exchange rates, both official and unofficial, at that time were around 3,000 krb per dollar, implying total GDP of \$4.6bn, or about \$92 per capita. This is an extraordinarily low number by any standards, but particularly considering that pre-independence purchasing power parity numbers suggested a per capita income around \$3,860.

In part, the problem is an inflation-induced real appreciation of the dollar — people buy dollars as a hedge against inflation and this tends to reduce the value of everything denominated in karbovantsi. But real output has also declined unequivocally.

The numbers measured in physical units are unambiguous. From the first quarter of 1992 to the first quarter of 1993, the output of electric energy fell 5 percent, the output of coal fell 5 percent, the output of chemicals fell 17 percent, the output of wood fell 3 percent, the output of pulp and paper fell 15 percent, the output of bread fell 11 percent, the output of meat fell 17 percent and the output of milk fell 35 percent. On top of previous declines, from the third quarter of 1992 to the first quarter of 1993, the output of ferrous metals fell by 15 percent, chemicals by 19 percent, bread by 20 percent.

Official statistics indicate that output fell by relatively less in Ukraine than in other former republics. Ukraine's decline in net material product was 15 percent in 1992, which was a larger fall than in Belarus (11 percent), Uzbekistan (12.9 percent) and Kazakhstan (14.2 percent), but less than in Russia (20 percent) and the others. Ukraine's 9 percent fall in gross industrial output in 1992 was less than in other former republics, with the exception of Belarus (PlanEcon 1993).

In contrast, inflation in Ukraine is the highest of any ex-republic, with the possible exception of some of those currently at war. Ukraine's retail price level rose almost 22 times from December 1991 to December 1992, an increase which was exceeded only by Azerbaijan. Ukraine's 25-fold increase in officially measured wholesale prices was the highest of any former republic.

Output has fallen sharply in Ukraine as it has in most other parts of the former Soviet Union and for similar reasons. Trade between former republics has been disrupted and the energy prices have risen steeply. What really sets Ukraine aside is its rapid rate of inflation, particularly over the past 12 months. Why has inflation risen so far and so fast?

III. PARTIAL MONETARY INDEPENDENCE, DECEMBER 1991 TO NOVEMBER 1992

Despite declaring independence in August 1991 and becoming formally independent after a referendum on December 1, borders between Ukraine and Russia stayed almost completely open and Ukraine remained in the Russian ruble zone — using so-called non-cash (*beznalichnye*) rubles for many transactions, particularly between enterprises, until November 1992.

As a result, the two economies remained closely linked, and when the Russian government announced that state controlled prices would be increased and "freed" at the beginning of 1992, the Ukrainian government felt the need to do likewise. However, Prime Minister Fokin and his colleagues declined to take further systematic reform measures. Ukraine during 1992 was characterized by the disintegration of state control over the economy, while the central bank provided low interest credits to favored sectors. "Gradualism" in Ukraine meant changing the previous structure of industry and trade as little as possible. At the same time, forms of spontaneous privatization swept through the state enterprise sector (Johnson and Kroll 1991).

In fall 1991, before full independence was established, our surveys in Kiev showed goods in state stores had stable prices although there were some shortages — seen in the lines for goods and in the periodic absence of some goods. For example, our data at this time — shown in Table 2 — indicate milk and bread were usually available, although with lines.⁴

Table 3 measures inflation by tracking the cost of a typical basket of goods. The basket of goods used in this calculation is the monthly ration of soldiers in the former Soviet (now Ukrainian) army. This ration comprises the following: 0.9 kg of butter, 1.5 kg sugar, 4.5 kg of meat, 12 kg bread, 2 kg cabbage, 2 kg of tomatoes, 2 kg of cucumbers, 9 kg of potatoes and 8 eggs. The ration also includes some less important goods for which we have not collected prices and excludes at least one important category of good — milk and other dairy products. In this price index, we use the state store price in all cases when a good is available in these stores, except when the line for a good is longer than 15 people, in which case we use the price in the "peasant market"

4 Opinions differ, but it is not uncommon to hear locals claim a real line (ochered) exists only if more than 10 people are waiting.

("kolkhoznii rinok" is the Russian term).⁵ This provides a crude measure of the way in which the shortages and waiting in line pushes up the cost of goods.

Table 3 shows significant inflation already in November and December 1991, when the price index increased 16 percent and 26 percent respectively. These price increases were almost exclusively due to increases in the price of goods in the peasant market. The major increases in state prices were still to come.

By the end of 1991 it was clear that the Ukrainian government intended to follow Russia at least partially and raise the prices of some goods at the beginning of January 1992. The government in Kiev also announced that it would introduce a new form of cash money, the coupon-karbovanets, at the beginning of the year. (Hereafter the coupon-karbovanets will be referred to as simply the coupon.)

The exchange rate of the coupon in its first days was high. On its first day, January 8, one coupon was worth 8 (Russian or Soviet) rubles, but within 2 weeks one coupon could only buy 2 rubles and by the end of February the coupon and ruble were trading one-for-one.

In effect, the government signalled clearly that prices would rise sharply in the New Year and the predictable result was that people bought all the goods they could find. By December 23, 1991, there was no bread, milk or eggs in any the stores which we survey. In fact, none of the 22 goods on our list was available in the surveyed stores on that day. During the last 2 weeks of December, most goods, including bread, were absent from the seven stores we survey, and the longest recorded line was 150 people waiting for milk.

There was a sharp anticipatory jump in market food prices at the end of 1991, with a large increase in the first and second weeks of December 1991 and in the first week of January 1992. The rate of growth was highest for meat products and for tomatoes, but all prices — with the curious exception of butter — increased by more than 200 percent. Table 3 shows that our measure of the retail price index increased more than 400 percent in January 1992. The goods situation did not improve immediately once prices were increased. In the first 2 weeks of January, no surveyed store had more than 8 types of goods out of the 22 goods on our list.

The price increases of January did not by themselves initiate an unstoppable inflation spiral. From mid-January to early March 1992, prices in state stores were held constant — with the exception of eggs — and prices in the peasant market actually fell.

5 Since independence, as usually in the past, the peasant market has offered people a wider range and higher quality of products than are available in state stores. However, prices are almost always higher than in state stores.

It appears that the extent of anticipatory purchases at the end of 1991 was so great that prices in the free market actually overshot their new equilibrium level. According to our data, from the first week of January to the first week of March there was a 22 percent fall in prices in the peasant market. Table 3 reflects this with a 6 percent fall in the retail price index in February and a 3 percent fall in March.

This should have been a major assistance to the government in holding down prices and Table 3 shows extremely good inflation results through June — for all months except April, which had an increase of 7 percent per month, i.e., an annualized rate of 125 percent. At the same time the black market exchange rate of the coupon stabilized against the ruble and held at around 120 coupons to the dollar from February to June — see Table 1. Unfortunately, government macroeconomic policy proved inconsistent with continued price stability.

The Ukrainian government's budget deficit was actually quite small in the first half of 1992, but grew steadily during the year. For the first 6 months the general budget deficit — for the central government plus regional authorities — was estimated at 42 billion coupons, which was around 2 percent of GDP. For the 9 months to the end of September this deficit had risen to 10 percent of GDP and for the whole year it was at least 16 percent of GDP (Popiel 1993a, p. 16).

The government's major problem in 1992 was the shrinking of revenue in real terms, rather than an increase in real spending. Although it is hard to measure real values correctly when inflation is so high, general government revenues in 1992 appear to have been around 27 percent of GDP and general government spending was around 43 percent. In contrast, government revenue was 33 percent of GDP in 1989, 34 percent in 1990 and 34.8 percent in 1991; government spending was 27.3 percent of GDP in 1989, 31.4 percent in 1990 and 48.3 percent in 1991. Real government spending actually appears to have fallen since independence.

Even more important, of course, is that any Ukrainian government deficit is financed almost exclusively by the printing of money. Further monetary emission through credits at low nominal interest rates have been used to effectively subsidize selected industrial sectors. The growth in nominal money from the end of 1991 to the end of 1992 was remarkable. The amount of currency in circulation rose 21 times, the value of M1 — currency and demand deposits — rose 10 times and the value of total monetary liabilities — M1 plus other deposits — also rose 10 times (Popiel 1993c, p. 41).

Currency was 13 percent of total monetary liabilities in December 1991 and 25 percent one year later. The share of household and enterprise demand deposits in total monetary liabilities fell from 27 percent to 7 percent and from 34 percent to 20 percent respectively. Calculated as a proportion of GDP, M1 was 60 percent and total monetary liabilities were 79 percent. There may have been some monetary

overhang, in the sense of money which could not buy goods in 1991, but this was almost certainly dissipated by the end of 1992. The M1/GDP ratio was down to 44 percent and the total monetary liabilities to GDP ratio was down to 57 percent. By way of contrast, in 1989 the ratio of broad money to GDP was 23 percent in Turkey, a country with about the same per capita income in purchasing power terms as Ukraine, 51 percent in Yugoslavia, a socialist country with high inflation, and 76 percent in France. There is still room for a further fall in the demand for real money balances in Ukraine. In a high inflation situation, lower real money demand usually leads directly to higher prices.

A rough indicator of broad money, without individuals' demand deposits and without other deposits indicates growth of 178 percent per quarter in the second quarter, 106 percent in the third quarter and 95 percent in the fourth quarter of 1992 (Popiel 1993a, p. 17). At the beginning of 1992 most monetary growth was the result of loans to state enterprises but the relative importance of this form of money creation declined as the budget deficit worsened.

Although it is hard to measure the real value of nominal magnitudes in a period of high inflation, the evidence suggests that the real budget deficit had risen dramatically by August 1992. Government spending was 20.2 billion coupons in January and 167 billion in August. Our price index, rebased to equal 100 in January, is 159 in August 1992. This implies real spending in August was 5 times higher than in January and that the deficit was 11 times larger in real terms than in January. In nominal terms the deficit was 3.2 billion coupons in January and 59 billion in August.

Published official statistics do not include any plausible numbers for money emission before the January-June period, so we cannot calculate, for example, monthly growth rates of money during spring 1992. However, for the January-June period about 100 billion coupons of the money growth was not due to the budget deficit, and so presumably entered the economy through low interest loans provided to state enterprises by the Ukrainian central bank — a key element of the Ukrainian government's "gradualist" economic policies.⁶ This fits with anecdotal evidence obtained from our interviews with managers of industrial firms and banks (Johnson, Kroll and Horton 1993).

6 A full explanation of the Ukrainian inflationary process would need data on the relative importance in money issued of cash coupons and *beznalichnye* rubles, and some estimate of how much of these *beznalichnye* rubles was spent in Russia. During 1992 the Ukrainian central bank could and did, in effect, write checks on the Russian central bank.

Also important is the time required to transfer funds from Ukraine to Russia and vice versa — in spring 1992 this was about 3 months, so payment orders in process probably represented a significant delayed inflationary push. Unfortunately, we do not have data on the amount of payment orders outstanding at any time.

Information about inflation in our price surveys during this period has to be interpreted with care, because there is strong evidence that during 1992 significant de facto price controls remained on some key goods — particularly milk and bread. The price of milk was not altered from January to June and the price of bread was held constant until July. How did this pricing policy affect the balance of supply and demand?

Although the line length for bread remained about constant from January to June, milk was seldom available until after its price was increased in June.⁷ For other less crucial goods sold in state stores, there appears to have been price regulation, but probably due more to the bureaucratic inertia of changing prices in state trading organizations.

For example, there were repeated changes in the price of eggs during spring 1992.⁸ Interestingly, these changes are hard to correlate with the evidence on changes in availability of eggs. After the price increases in January eggs were briefly available in state stores, although with a line. The price cuts of February seem to have resulted in eggs disappearing from the state stores, at least until April, from which date they were available with a line — although the price of eggs was not increased until June. The most plausible explanation for these movements is that the state trading organizations dealing with eggs were trying to achieve rough balance in the market and had a great deal of difficulty getting the price right.

Bread and milk exhibit more price stability than most other goods sold in state stores during this period — an indication that these are viewed by the government as strategic goods. A further strong indicator of price controls is that, in spite of the gathering inflation momentum and already significant lines, the price of milk was reduced in mid-September. As Table 2 shows, milk disappeared from state stores until much higher prices were introduced in February 1993.

The most important conclusion from price behavior and the availability of goods is that while price controls had been loosened, prices in state stores were not now "free." Through 1992 and still in summer 1993, prices continued to be set by administrators, either in the stores themselves or — more usually — in the relevant state wholesale trade

7 It is strange that although bread was available with only a short line until the summer, line lengths actually increased after bread prices were raised in July. One possible explanation is that the general rise in food prices in summer 1992 pushed down real incomes such that poor people, particularly pensioners, actually bought and consumed more bread than before. It is also possible that expectations of rising inflation induced people to buy and store bread — although in warm weather the ability to store bread is limited by people's freezer capacity.

8 One significant difference between eggs and milk or bread is that when eggs disappear from state stores they can — and often do — appear in the peasant market.

organization. These administrators respond to price and line length signals only with long and variable lags.

As a result, prices in state stores exhibited considerable inertia and should be considered as lagging behind the overall inflation of the economy.⁹ The leading indicator of Ukrainian inflation was usually the black market exchange rate of the coupon against the dollar, while prices in the peasant market provide a further indicator of current and future inflation. Our retail food price index, reported in Table 3, provides a summary measure of current inflation.

The second half of 1992, particularly from August, was characterized by the rapidly accelerating depreciation of the Ukrainian currency. The coupon remained basically stable against the dollar from February to June 1992, but during the summer it began to depreciate significantly. As Table 1 shows, from the beginning of September 1992 the rate depreciated steadily, moving at least 5 coupons every 2 days. The more rapid depreciation at the beginning of October was due to the effect on inflation of higher administered prices — particularly gasoline — at the beginning of the month. In the last two trading days of October and the first days of November, the exchange rate depreciated 5 coupons a day against the dollar for the first time.

The first 9 months of Ukrainian independence shows the effects of partially liberalizing prices while refusing to speed the privatization of trade. In Ukraine, the "liberalizing" of prices meant increasing state controlled basic prices — particularly for energy — removing some subsidies at the level of intermediate goods, and decentralizing price setting authority to the level of the bureaucrats running state trade organizations. There was some growth of private trade, particularly in nonfood products, but almost all shops and wholesale trade organizations remained just as much in the state sector as before. The only exception, at least in Kiev, was a few shops which were now leased but which usually offered very similar goods and services as before. By early November 1992, on the verge of full monetary independence, the Ukrainian budget was out of control, credit was loose and prices were rising sharply.

IV. FULL MONETARY INDEPENDENCE

From January 1992 the role of Russian rubles in cash transactions in Ukraine had been gradually displaced by the karbovanets (Ukrainian for ruble and popularly known as the coupon.) By November 1992 it was common to use coupons for almost all retail purchases, although rubles could be used in some instances at the current black market rate of exchange and Russian coins (kopecks) were used in those increasingly rare instances when change was needed.

9 It may also be the case that the costs of key goods rise while their prices are held constant and the government directly or indirectly finances the difference. While this mechanism seems plausible even in summer 1993, the supportive evidence is unfortunately only anecdotal.

At the beginning of October 1992, the Russian central bank stopped honoring at least some of the payments authorized — in Russian rubles — by the Ukrainian central bank. The Russian government subsequently made it clear, most notably at the CIS meeting in Bishkek, that ex-republics should either abide by Russian monetary rules or get out of the ruble zone. In large part this was a response to the large amounts of credit issued by the Ukrainian National Bank.

On November 12, 1992, the government in Kiev reacted to this pressure by leaving the ruble zone and introducing a non-cash (*beznalichnye* in Russian) coupon. It also declared that the Russian ruble was no longer legal tender in Ukraine. Existing balances in Ukrainian banks were converted from non-cash rubles to non-cash coupons at an exchange rate of one for one. For all new transactions, however, the *beznalichnye* coupon-ruble exchange rate was fixed by the Ukrainian National Bank at 1 ruble to 1.45 coupons.

This initial official rate was fairly close to the black market rate — on November 12-14 our survey showed the cash rate, for buying coupons, to be 1.4 coupons per ruble. However, the rate moved to 1.5 on November 15, held at 1.55 from November 16 to November 21, and then began the fairly steady depreciation shown in Table 1. This was an early indication that Ukrainian monetary independence would mean more rather than less inflation compared to Russia.

The move to full monetary independence was followed by strong pro-stabilization announcements by the government. For example, on November 18, 1992, Prime Minister Kuchma addressed the parliament, with the supposed aim of establishing what anti-crisis measures were required (*Golos Ukraini*, November 19, No. 221)

On the basis of these warnings, over the next month the parliament was persuaded that the government should be granted strong anti-crisis powers. From December 1992 to May 1993 the government was empowered to rule by decree on economic matters, subject to confirmation by parliament. In practice, it appears that the government had a great deal of discretion over economic policy during this period.

There were some initial signs of fiscal austerity but these soon faded. In January there was a budget surplus amounting to 48 percent of government spending, but this may have been due to the immediate effect of higher administered prices on the nominal value of state enterprise sales and thus on the corporate income taxes. In February the deficit was already 11 percent of spending and it rose to 14 percent in March and 65 percent in April. For the whole period January-April, the budget deficit was 21 percent of total government spending (Popiel 1993c, p. 40).

Monetary emission also continued to be rapid. M3 growth was 31 percent in November 1992, 26 percent in December, 29 percent in January 1993, 17 percent in February and 46 percent in March. The government subsequently claimed that the

central bank had granted far more in credits to state enterprises than had been authorized, but it seems plausible that some government ministers knew what was going on. The government's emergency powers gave it greater control over the central bank than previously, but at the very least this control was not exercised effectively.

Unfortunately, therefore, the government's extra powers did not result in anything approaching a coherent anti-inflation policy and monetary emission has continued unchecked. It is not presently possible for outsiders to unravel the precise allocation of blame for this lack of stabilization policy, but the outcome in terms of inflation is painfully clear.

The October and November rates of depreciation were in the 30-40 percent range, but this fell into the mid-20 percent range in early December and down below 20 percent in the last 2 weeks of December. However, a new round of administrative price increases in December 1992 led to more rapid depreciation through early February, peaking at 90 percent for the month to February 8. The following month showed a relative deceleration in the rate of depreciation, with 2,000-2,200 coupons to the dollar appearing to be a resistance level — in the sense that traders were unwilling for a time to move beyond it. The rate moved fairly rapidly towards 3,000 coupons to the dollar during April, and then again displayed relative stability — holding in the range of 3,000 to 3,100 from early May to early June.

Table 3 shows the dramatic surge in Ukrainian food prices in December 1992, with the index rising more than 250 percent. The scale of Ukrainian price increases in December 1992 is quite evident. The price of bread doubled from 16 coupons per kilo in the second week to 32 coupons per kilo in the fourth week. This jump increase was followed by a steady series of increases, often followed by a couple of weeks with a stable price. Inflation fell in subsequent months, but remained in the range of 10-15 percent.

In the case of bread, the line length remained roughly constant and consistently less than 10 from October through mid-April 1993. Evidence of worsening disequilibrium in the bread market is given by the very long lines at the end of April and in May. These were the longest bread lines at least since our surveys began in fall 1991 and clearly indicated that the ordinary sequence of bureaucratic price raising was not enough to clear the market. In response, bread prices were tripled between the first and second weeks of June.

The supply of meat has been similarly erratic in Kiev state stores. In 4 out of the 11 weeks in 1993 for which we have data, no meat was available. In 3 other weeks the line length was 15 or more people. Inflation data for goods sold in kolkhoz markets shows large fluctuations in relative prices every month and that many prices rise steeply and then fall in the aftermath of increases in administrative prices. For example, in Kiev

during January-March, 5 out of 14 goods experienced price declines and another three had one month without any increase in price.

On June 1, the Ukrainian government announced that subsidies would shortly be withdrawn from meat, milk and bread. The predictable result was a sharp increase in kolkhoz market prices. Pork rose from 3,000 coupons per kilo on May 30 to 7,000 on June 8, potatoes increased from 80 coupons per kilo to 200 and the price of onions moved from 100 coupons per kilo to 250 coupons.

Also predictable was a wave of panic buying. There were big increases in bread and milk lines in state stores. Milk was only present in one of the three stores which we surveyed, there was no meat in any of our surveyed stores and the line for eggs was the longest it had been all year.

The announcement of official price increases at the beginning of June had a direct effect on the black market exchange rate, which fell 19 percent from June 2 to June 8. The rate may have initially overshot its new equilibrium value — on June 11 it had risen back to 3,400 coupons to the dollar — and the unsettled nature of the black market is apparent in the fact that no traders were willing to sell dollars during our surveys on June 8 and June 11.

We have yet to obtain the latest monetary statistics, but a great deal of new money has been created in the past few months. A miners' strike was ended in June in part through promising extra credits and other forms of financial assistance estimated to cost between 7 and 13 trillion coupons — although the whole broad money supply was only 6.5 trillion coupons in April 1993. Additional credits of around 2 trillion coupons for agriculture and state enterprises were also approved in June.

From the end of May to the end of June 1993 there was only a 4 percent increase in currency in circulation but a 73 percent increase in state enterprise deposits. This suggests the renewed importance of credit creation in the monetary emission. The net result for June was that M1 rose by 43 percent and M3 by 47 percent — the highest rates since monetary independence.

The acceleration of the rate of money creation presumably explains the rapid rise in the black market exchange rate from 3300 coupons to the dollar on July 12 to 4,300 on July 17. In recent weeks, the rate has dropped dramatically: to 4,500 on July 26, 4,850 on August 4, 5,700 on August 11 and 6,200 in the afternoon of August 13.

The Ukrainian food supply situation remains approximately the same as it was just before independence. The supply of food in state stores is erratic with frequent periods of long lines and periodic disappearance of goods. The quality of state products remains poor. At the same time inflation has accelerated dramatically, is now higher than in Russia and is moving towards hyperinflationary levels. The exchange rate has begun to

depreciate by as much as 7 percent a day and the fundamentals of monetary and fiscal policy appear very poor.

On top of these problems must be added the recent miners' strike. Miners are particularly important because they are the best organized group in the predominantly Russian area of the Donbass. This area voted overwhelmingly for independence in December 1991 — for example, both the Donetsk and Luhansk oblasts had 83.9 percent in favor — in part probably because they thought living standards would be higher than if they stayed under the jurisdiction of Moscow. The recent strike indicates that eastern Ukraine is very dissatisfied with current economic and political situation. If the economy continues to perform better in Russia than in Ukraine, dissatisfaction will probably grow. While open inter-ethnic conflict or secession is unlikely in the near future, this is a destabilizing factor.

Ironically, it appears that by breaking its direct link with the ruble, the Ukrainian government eliminated the last nominal anchor in the economy. The willingness of the Russian central bank to finance Ukraine may have been a very imperfect anchor previously, but it is now gone. The coupon is a money with no credible prospect of stabilization.

V. THE FAILURE OF PRIVATIZATION

In Ukraine, as in most post-communist countries, privatization is at least nominally at the top of the political agenda (Frydman et al 1993) and the Ukrainian Privatization Program, adopted by parliament in July 1992, lays out an ambitious timetable for achieving an impressive set of goals (State Program 1992). But so far there has been little progress.

Six criteria are commonly used to evaluate privatization proposals in post-communist countries and to define what it means for these programs to be successful. The first four criteria are concerned with the process of privatization, while the fifth and sixth focus on the nature of firms which privatization creates.

First, privatization should be rapid. In particular, sufficient privatization needs to take place while there is still a clear political consensus in favor of privatization. On this point Ukraine has a very disappointing record. The only privatizations to date have been relatively few shops in a couple of towns and a couple of initial public offerings organized by the International Finance Corporation. In contrast Russia has proceeded much faster, both with shops and with the privatization of larger enterprises. Political opposition to privatization appears to be growing in Ukraine and the legislature has still not passed the 1993 Privatization Program.

Second, privatization should be complete, in the sense that it results in the transfer of all de jure property rights to private persons. If the privatization is initially

incomplete, for example by only allowing leases, there needs to be a clear procedure and timetable for private persons to acquire all de jure rights. There have been several attempts to clarify the status of leased enterprises in Ukraine, but in practice the situation remains somewhat murky.

Third, privatization should be clean, in the sense that it does not involve too much corruption. Ukraine has not yet progressed far enough for a full assessment of this point, but there are reports that up to half of the purchasers of privatized shops have been unable to make payment in full. If there has been any corruption of local officials this will presumably have negative repercussions on other privatization projects.

Fourth, the privatization process needs to allow sufficient opportunities for all interested citizens to participate. As in many other post-communist countries, the Ukrainian government decided that a form of privatization voucher would broaden participation in privatization. Unfortunately, the initial scheme adopted was rather too complicated and proved largely unworkable (Johnson and Eder 1992). These problems delayed the privatization of larger enterprises and have yet to be completely resolved.

Fifth, privatization should create firms which have effective corporate governance, so stockholders are able to effectively supervise managers and to prevent them from taking actions which are not in the best interests of the stockholders. Sixth, privatized firms should be structured in a way which allows them to bring in outside capital, either in the form of debt or equity. It remains far from clear how these two criteria will be met in Ukraine.

In the absence of a coherent government privatization strategy, insiders — workers and managers in state firms — have acquired some property rights for free or at very low prices. However, they do not own the whole firm, so privatization remains incomplete, and — even worse — there does not currently exist a clear set of rules for converting partially privatized assets into fully privatized assets. Nevertheless, insiders have obtained strong claims to be able to minimize the property rights obtained by outsiders, so there will be only limited participation by the population as a whole.

A decree on the conversion of Ukrainian state enterprises into joint stock companies was signed by President Kravchuk on June 15, 1993, and it confers a strong role on ministries and other state agencies. Whether they can effectively control managers and the process of privatization remains to be seen.

The current situation is firms in which managers and workers have considerable de facto property rights, but almost never full de jure ownership. They have control over the property, but it is not be their property. Threats by the government to take this property away from them presumably reduce their time horizon and make them less likely to invest and more likely to clandestinely transfer assets to the private sector.

These firms do not have a proper system of corporate governance and they find it very difficult to bring in outside capital.

VI. THE BEGINNING OF STRUCTURAL CHANGE

Despite the poor overall performance of the Ukrainian economy, there is some evidence of positive changes. Our own observations and anecdotal evidence strongly suggest that the availability and quality of nonfood goods continues to increase in Ukraine. However, we have not been able to develop any relevant systematic measures, primarily because many of these goods are now distributed through new retail outlets. For example, a great deal of private nonfood trade in Kiev still takes place in temporary stalls at the Republican Stadium, but as traders accumulate capital they acquire premises and usually try to specialize.

There have been two developments in nonfood trade since independence. First, in 1991 anecdotal evidence suggests most goods on sale were purchased in the former Soviet Union — many of them at low, state controlled prices — although there were some imports from Poland. From fall 1992 more goods were directly from Turkey, China, Hong Kong and Taiwan. Second, there are now more goods on sale and the range of choice is wider. Our very rough numbers on prices indicate rates of inflation in line with those given above for food products. Unfortunately, we have not been able to construct a meaningful price index for these goods.

Interestingly, while private business now appears quite active in nonfood trade, there has been relatively little entry by private business into the wholesale food trade. The retail food trade remains just as divided as before — products can either be bought in state stores or in the peasant market.

More broadly, the private sector has begun to show sustained growth over the past year. At this time most of the growth is still in trade and simple services, but this is the start of significant job creation outside the state sector. In a survey of 349 people who were fired from state enterprises in the second half of 1992, we found that by December 1992, 42 percent had found work in the nonstate sector and 31 percent had found a new job in the state sector (Johnson and Ustenko 1993c). These rates of reemployment are very high and it remains to be seen if they will be maintained.

There has also been rapid growth in the scale and employment of new commercial banks — where "new" indicates that they were not formed out of the former Gosbank system. Based on a survey conducted in February 1993, we estimate that the approximately 130 new commercial banks in Ukraine have own capital and deposits close to that in the four largest former state banks. The difference in size of operations between these two parts of the banking system is accounted for almost entirely by credits provided by the central bank to firms, because most of these pass through the former state banks.

Furthermore, we also found evidence that at least some of these new commercial banks have moved beyond being narrowly based on just one sector. In their efforts to raise more capital and to diversify their risks, successful banks are seeking more shareholders and a broader set of customers (Johnson and Ustenko 1993a).

The evidence so far on the agricultural sector suggests that directors of state and collective farms have been quite effective in slowing any kind of meaningful transformation. The large state farms continue to coexist alongside the more efficient private plots of rural and urban residents. At the same time, there are now some indications of an increase in individual farming oriented towards market production rather than self-sufficiency (Johnson and Minton Beddoes 1993).

The extent of serious adjustment within state enterprises remains unclear. In the absence of effective government-directed privatization there continues to be some spontaneous privatization, more in the form of assets leaking out of the state sector than complete transformation of legal property forms. The available evidence suggests that these steps by themselves have not significantly improved the performance of state enterprises and the outlook for jobs in the state sector remains poor (Johnson and Ustenko 1993b and 1993c).

Very high inflation usually reduces the real tax revenues of government, unless taxes have been effectively indexed. Not only is such indexation absent from Ukraine, but state enterprises are particularly suited to igniting inflation. These enterprises have no real owners (leaving aside the property rights claimed and exercised by managers and workers themselves), cannot easily go bankrupt and are not now subject to effective supervision by the tax authorities. Loans have been available at negative real interest rates, there are few effective price controls and real wages have already fallen a great deal. As early as 1991, our interviews with managers suggested they were beginning to find ways to reduce their real tax liabilities. Real tax revenue from enterprises fell from 11-13 percent of GDP in 1989-91 to around 6 percent in 1992.

In short, while there are some signs of structural change in the Ukrainian economy, progress so far has been limited. The government has certainly not demonstrated a firm commitment to development of the private sector, let alone a coherent strategy of promoting structural change. The acceleration of inflation appears to have distorted and delayed the most positive changes. The lack of privatization makes it harder to stabilize because state enterprise managers lobby hard for various forms of financial assistance.

VII. WAGES AND LIVING STANDARDS

We assess the impact on real wages by comparing nominal wages to the amount needed to buy the basic basket of food for which we report prices in Table 3. In October 1991, of the goods in the "Ukrainian Army Ration" (Table 3) only bread and sugar were

available in state stores, while the rest had to be purchased in the market. The whole basket of goods would have cost 198.2 rubles. This represented 50 percent of average wages. (This is using an average wage of 400 rubles, based on our interviews at this time.)

In contrast, in Kiev at the beginning of October 1992, the same basket of goods cost 3,067 rubles — 123 percent of average wages. (This uses an estimate of average wages in Ukraine which our interviews suggest is on the high side: 2,500 coupons.) Even though there is a considerable margin of error in these numbers, this is strong evidence that average real wages fell during 1992.

People working in the state sector in November 1992 had average wages which were 2-3 times what was required to buy this basket, while the ratio in the relatively high wage mining sector was 10 times and in the relatively low wage retail sector it was 1. By March 1993 these multiples for state employees had fallen to 1 for average wages, 2 for miners and 0.5 for the retail sector. In late June 1993, following their strike, miners had increased their multiple to 5, while average wages had fallen to 0.6 and retail wages had dropped to 0.3.

Our assessment that average real wages have fallen is also strengthened if we go beyond merely food items and consider what people have to pay for their apartments. The basic cost for an apartment in Ukraine depends on the number of square meters in the apartment and on whether there is a telephone. One payment — referred to in Russian as "payment for the apartment" ("zaplata za kvateri.") — covers most costs, including rent, water and gas. (Electricity is metered separately and telephone costs depend on use.) These costs rose at least 10 times during 1992 and have increased even more rapidly so far in 1993. In addition, the cost of a monthly pass which allows the use of all types of municipal transport — trolleybus, tram, bus and metro — has risen massively in price. We estimate that apartment payments and transportation costs were no more than 6 percent of average wages in 1991 and now constitute at least 10 percent.

In addition, for at least one section of the population living standards remain unambiguously at disastrous levels. In fall 1991 pensioners received 130 rubles, 66 percent of the cost of our basket of goods. At this time the living standard of pensioners had already dropped steeply and was widely regarded as being too low. By the beginning of October 1992, the average pension was around 1,100 coupons, 36 percent of the basket's cost. The purchasing power of pensions has dropped still further in 1993.

An important caveat must be attached to this assessment because a great deal depends on the collective earnings of the family unit which shares an apartment. For example, it matters a great deal whether a pensioner lives with someone who earns at least an average salary. Unfortunately, a fuller assessment of living standards requires

data on the composition of households and the intra-household sharing of income which we do not currently have.

Overall, real living standards have fallen for most social groups since independence. The exception is probably the relatively few people who have managed to enter high-paying parts of the private sector. For the vast majority of the population the progress towards hyperinflation has meant steady impoverishment.

VIII. CONCLUSION

Independent Ukraine has faced significant shocks to its economy, particularly in the form of disrupted trade with Russia and higher energy prices. These shocks are smaller in relative magnitude for Ukraine than for most former Soviet republics, yet Ukraine's macroeconomic performance has been just as bad on most measures, its inflation has been worse and its progress in creating a genuine private sector has been slower. The collapse of the Ukrainian coupon against both the dollar and the ruble since full monetary independence at the end of 1992 underlines that the major problem in Ukraine has been the government's economic policy.

The economic strategy of Ukraine since independence has been a form of gradualism, implemented by ex-communists who realize that something called "economic reform" is required but who are not willing to take the decisive measures. This same strategy has been followed by both the Fokin and Kuchma governments, although the latter claims to be in favor of radical reform. But the Kuchma government has not acted decisively on the budget, credit, privatization or liberalization.

Even in comparison with the yet incomplete reforms in Russia, Ukraine fares badly. Ukraine has undoubtedly fallen behind Russia in terms of initiating privatization and liberalizing internal trade. Our research suggests that these measures already mean better availability of goods, particularly food, in at least some parts of Russia compared with Ukraine (Johnson and Ustenko 1993d).

It is true that Russia has also failed to bring inflation completely under control but the latest data suggests an acceleration of inflation in Ukraine but not in Russia. Russia has moved further in terms of reforming public finance and negotiating meaningful economic assistance from the West. The West appears willing to help Ukraine, despite the current ambiguities of Ukraine's position on nuclear weapons, but Ukraine does not yet have an economic policy to which western assistance would make a significant difference.

There are some positive signs in terms of new economic developments, particularly outside the state sector. However, the continuation of inflation cannot be considered helpful to these new forms of business. High inflation tends to keep horizons short, to make it harder to build long-term relationships, and to cause a great deal of

volatility in relative prices. Full-fledged hyperinflation could cause significant damage to the emergent private sector.

In an economy without long experience of inflation and without smoothly functioning indexation mechanisms, there is a tendency for very high inflation to become hyperinflation, involving sustained price increases over 50 percent per month. Our evidence indicates Ukraine is slipping now into hyperinflation and unless dramatic stabilization measures are taken immediately, full-fledged hyperinflation will emerge in the fall.

The required stabilization measures comprise much more than the partial measure of increasing some administered prices. As our evidence on the food market shows, the primary result of administered price increases is at best temporary improvements in the availability of goods. The advice of western organizations is almost unanimous: only a set of measures which closes the budget deficit and imposes strict limits on credit can have any chance of stemming inflation at this point. Supportive measures in the form of trade liberalization and full currency convertibility would also be very helpful.

Is there any hope? In our assessment, recent experience in Eastern Europe shows three necessary conditions for economic reform: politicians who can maintain sufficient support to implement economic policies which inflict costs on part of the population, bureaucrats who cannot obstruct the required changes and enough competent economic policymakers.

The comparison with Poland in 1989 is instructive, because Poland was also facing runaway inflation due to fiscal deficits and loose credit. The situation began to improve only after the appointment of Leszek Balcerowicz as Minister of Finance. The key conditions for stabilization and the start of real transformation of the economy were the abilities of Balcerowicz and his team, and the political courage of the leaders of Solidarity (Johnson and Kowalska 1993).

Ukraine has few people with sufficient knowledge and experience to make economic policy, but it has enough. Its bureaucrats are weak and do not present the kind of obstacles now apparent in Russia. But independent Ukraine lacks political leaders who understand why and how inflation must be controlled.

TABLE 1 BLACK MARKET EXCHANGE RATES IN KIEV, COUPON/DOLLAR AND COUPON/RUBLE EXCHANGE RATES				
	Trade Buying Dollars	Trader Selling Dollars	Trader Buying Rubles	Trader Selling Rubles
January 1992	120	--	--	--
February 1992	120	--	--	--
June 1	120	--	--	--
July 6	200	--	--	--
August 4	250	--	--	--
September 4	280	--	--	--
September 8	325	355	--	--
September 17	340	370	--	--
September 26	380	390	--	--
October 1	390	415	--	--
October 8	450 (39%)	490	--	--
October 17	480 (41%)	500	--	--
October 26	530 (40%)	555	--	--
November 2	560 (44%)	580	--	--
November 8	600 (33%)	620	--	--

TABLE 1 (CONT'D) BLACK MARKET EXCHANGE RATES IN KIEV, COUPON/DOLLAR AND COUPON/RUBLE EXCHANGE RATES				
	Trade Buying Dollars	Trader Selling Dollars	Trader Buying Rubles	Trader Selling Rubles
November 17	710 (48 %)	740	1.55	--
November 26	755 (42 %)	805	1.65	--
December 2	720 (29 %)	n.t.	1.90	--
December 8	740 (23 %)	n.t.	1.90	--
December 17	770 (8 %)	n.t.	1.85	--
December 26	860 (14 %)	930	1.80	1.9
January 2	1,000 (39 %)	1,100	1.95	2.0
January 8	1,050 (42 %)	1,150	1.95	2.0
January 17	1,200 (56 %)	1,400	2.20	2.3
January 26	1,470 (71 %)	1,570	2.50	2.7
February 2	1,600 (60 %)	1,700	2.70	2.75
February 8	2,000 (90 %)	2,150	2.80	3.0

TABLE 1 (CONT'D) BLACK MARKET EXCHANGE RATES IN KIEV, COUPON/DOLLAR AND COUPON/RUBLE EXCHANGE RATES				
	Trade Buying Dollars	Trader Selling Dollars	Trader Buying Rubles	Trader Selling Rubles
February 17	1,900 (58%)	2,000	2.8	2.9
February 26	2,000 (36%)	2,100	2.75	2.85
March 2	2,000 (25%)	2,150	2.75	2.85
March 8	2,000 (0%)	2,200	2.6	2.8
March 17	2,200 (16%)	n.t.	2.8	2.85
March 26	2,200 (10%)	2,300	3.0	3.2
April 2	2,200 (10%)	2,300	3.0	3.2
April 8	2,800 (40%)	3,100	3.0	3.2
April 17	2,750 (25%)	3,100	3.5	3.6
April 26	3,000 (36%)	3,100	3.2	3.5
May 2	3,000 (36)	3,150	3.2	3.4
May 8	3,000 (7%)	3,150	3.1	3.3

TABLE 1 (CONT'D) BLACK MARKET EXCHANGE RATES IN KIEV, COUPON/DOLLAR AND COUPON/RUBLE EXCHANGE RATES				
	Trade Buying Dollars	Trader Selling Dollars	Trader Buying Rubles	Trader Selling Rubles
May 17	3,050 (11%)	3,200	3.0	3.2
May 26	3,050 (2%)	3,200	2.9	3.1
June 2	3,100 (3%)	3,200	2.6	2.8
June 8	3,700 (23%)	n.t.	2.6	2.8
June 17	3,400 (12%)	3,550	2.6	2.7
June 26	3,450 (13%)	3,600	2.6	2.7
June 28	3,450	3,600	2.6	2.7
June 29	3,450	3,600	2.6	2.7
June 30	3,450	3,600	2.8	3.0
July 1	3,450	3,600	2.9	3.15
July 12	3,300	3,600	2.6	2.7
July 17	4,300 (27%)	4,100	3.4	3.6

TABLE 1 (CONT'D) BLACK MARKET EXCHANGE RATES IN KIEV, COUPON/DOLLAR AND COUPON/RUBLE EXCHANGE RATES				
	Trade Buying Dollars	Trader Selling Dollars	Trader Buying Rubles	Trader Selling Rubles
July 21	4,450	4,700	4.1	4.2
July 26	4,500 (30%)	4,700	4.2	4.3
August 4	4,850	5,200	4.4	4.6
August 10	5,300	5,800	4.7	5.0
August 11	5,700 (33%)	6,200	4.8	5.2
August 12	6,000	6,500	4.9	5.5
August 13 Morning	6,100	6,700	4.9	5.5
Afternoon	6,200	6,900	4.9	5.5
"n.t." indicates there were no transactions of this kind. A blank denotes we did not collect information on this date. The number in brackets is the percentage change in the black market exchange rate compared with the same date in the previous month. Source: Inquiries in the same place at approximately the same time every day. Five people are asked for their selling rates and five people are asked for their buying rates. The numbers reported here are the mean prices after discarding implausible outlying quotes.				

TABLE 2 PRICES AND LINE LENGTHS FOR BREAD AND MILK, STATE STORES IN KIEV AND ST. PETERSBURG July 1991 - June 1993				
	Bread (kilo)		Milk (Liter)	
	Price	Line Length	Price	Line Length
July 1991				
23rd-24th	0.96	6	0.6	7
30th	0.96	5	0.6	10
August				
4th Week	0.96	3	0.6	12
September				
2nd Week	0.96	7	0.6	10
4th Week	0.96	4	0.6	5
October				
1st Week	0.96	5	0.6	5
3rd Week	0.96	4	0.6	3
November				
4th Week	0.96	5	0.6	n.p.
December				
2nd Week	0.96	10	0.6	n.p.
4th Week	0.96	n.p.	0.6	n.p.
January 1992				
2nd Week	5	15	1.95	60
4th Week	5	7	1.95	n.p.
February				
2nd Week	5	6	1.95	n.p.
4th Week	5	5	1.95	n.p.
March				
2nd Week	5	3	1.95	n.p.
4th Week	5	3	1.95	n.p.

TABLE 2 (CONT'D) PRICES AND LINE LENGTHS FOR BREAD AND MILK, STATE STORES IN KIEV AND ST. PETERSBURG July 1991 - June 1993				
	Bread (kilo)		Milk (Liter)	
	Price	Line Length	Price	Line Length
April				
1st week	5	10	1.95	n.p.
3rd week	5	5	1.95	6
May				
2nd Week	5	9	1.95	n.p.
4th Week	5	7	1.95	n.p.
June				
2nd Week	5	7	1.95	8
4th Week	5	9	18	10
July				
1st Week	16	10	18	10
3rd Week	16	15	18	15
August				
2nd Week	16	7	18	7
4th Week	16	15	18	15
September				
2nd week	16	n.p.	21	4
4th week	16	10	12	5
October				
2nd Week	16	4	n.k.	n.p.
4th Week	16	8	n.k.	n.p.
November				
2nd Week	16	4	n.k.	n.p.
4th Week	16	8	n.k.	n.p.

TABLE 2 (CONT'D) PRICES AND LINE LENGTHS FOR BREAD AND MILK, STATE STORES IN KIEV AND ST. PETERSBURG July 1991 - June 1993				
	Bread (kilo)		Milk (Liter)	
	Price	Line Length	Price	Line Length
December				
2nd Week	16	5	n.k.	n.p.
4th Week	32	4	n.k.	n.p.
January 1993				
2nd Week	40	3	n.k.	n.p.
4th Week	53	4	n.k.	n.p.
February				
2nd Week	65	8	178	10
4th Week	65	7	178	5
March				
2nd Week	65	4	n.k.	n.p.
4th Week	72	5	178	3
April				
2nd week	75	4	180	8
4th week	65	40	180	15
May				
2nd Week	60	30	180	17
4th Week	60	16	200	20
June				
1st Week	60	23	217	45
2nd Week	180	9	440	7
3rd Week	200	10	325	15
4th Week	210	8	325	17

TABLE 3 RETAIL FOOD PRICE INDEX (The Ukrainian Army Ration)			
Date	Cost of Basket	Cumulative Index	Monthly Inflation (Percent)
October 1991	198	100	---
November	230	116	16
December	290	146	26
January 1992	1,551	782	435
February	1,455	734	-6
March	1,405	708	-3
April	1,507	760	7
May	1,520	766	1
June	1,537	775	1
July	1,930	973	26
August	2,460	1,241	27
September	2,500	1,261	2
October 1992	3,150	1,589	26
November	3,543	1,788	13
December	12,709	6,412	259

TABLE 3 RETAIL FOOD PRICE INDEX (The Ukrainian Army Ration)			
Date	Cost of Basket	Cumulative Index	Monthly Inflation (Percent)
January 1993	17,968	9,066	41
February	20,688	10,438	15
March	23,887	12,052	15
April	26,820	13,532	12
May	32,850	16,574	23
June	69,560	35,096	112
July	84,000	42,381	21
August (1st Week Only)	93,000	46,969	11

Each month's index is computed using data from the second week of that month only. Monthly inflation is calculated by comparing the second week of the relevant month with the second week of the previous month. However, for June 1993 we also report data from the first week and fourth weeks.

Source: Our own price surveys. The composition of the basket is described in the text.

Until December 1991, the "cost of basket" is in rubles; from January 1992 it is in coupons.

REFERENCES

- The Economist 1993a. "You'd be nervous living next to a bear." May 15, pp.21-23.
- The Economist 1993b. "The post-cold-war war." June 19, pp.49-50.
- Frydman, Roman, Andrzej Rapaczynski, John S. Earle et al editors 1993. "Ukraine," in The Privatization Process in Russia, Ukraine and the Baltic States, Central European University Press, London, 1993.
- Hogan, William W. 1991. "Economic Reforms in the Sovereign States of the Former Soviet Union," in Brookings Papers on Economic Activity, vol. 2, pp.303-319.
- International Monetary Fund, The World Bank, Organization for Economic Co-operation and Development, European Bank for Reconstruction and Development 1991. A Study of the Soviet Economy, 3 Volumes, February.
- Johnson, Simon and Heidi Kroll 1991, "Managerial Strategies for Spontaneous Privatization," Soviet Economy, vol.7, no. 4, pp.281-316.
- Johnson, Simon and Santiago Eder 1992, "Prospects for Privatization in Ukraine," RFE/RL Research Report, vol.1, no.37, 18 September.
- Johnson, Simon and Marzena Kowalska 1993, "The Transformation of Poland, 1989-92," forthcoming in Voting for Reform, edited by Steven Webb and Stephan Haggard, 1994.
- Johnson, Simon, Heidi Kroll and Mark Horton 1993, "New Commercial Banks in the Former Soviet Union. How Do They Operate?" In Changing the Economic System in Russia, pp.183-209, edited by Anders Aslund and Richard Layard, Pinter Publishers, London.
- Johnson, Simon and Oleg Ustenko 1992. "Ukraine on the Brink of Hyperinflation," RFE/RL Research Report, vol.1, no.50, 18 December.
- Johnson, Simon and Zanny Minton Beddoes 1993. "The Acquisition of Private Property Rights in Ukrainian Agriculture," April, forthcoming in the proceedings of a conference held at U.C. San Diego.
- Johnson, Simon and Oleg Ustenko, 1993a. "The Role of New Banks in a Partially Reformed Economy: Ukrainian Experience," April, unpublished paper.

- Johnson, Simon and Oleg Ustenko 1993b. "Corporate Control of Enterprises Before Privatization: The Effects of Spontaneous Privatization," May 1993, forthcoming in the proceedings of a conference organized by the Kiel Institute of World Economics.
- Johnson, Simon and Oleg Ustenko 1993c. "Unemployment After Communism: Five Results from Ukraine," June, unpublished paper presented at the NBER summer workshop on labor economics.
- Johnson, Simon and Oleg Ustenko 1993d, "Ukraine Slips into Hyperinflation," RFE/RI Research Report, vol.2, no.26, 25 June.
- Koropeckyj, I.S. 1992. The Ukrainian Economy: Achievements, Problems, Challenges, Harvard University Press, Cambridge.
- Mearsheimer, John J. 1993. "The Case for a Ukrainian Nuclear Deterrent." Foreign Affairs, Summer.
- The New York Times 1993. "Belarus, a Model for Ukraine." Editorial, p.A14, Wednesday, August 4.
- PlanEcon 1993. "Ukrainian Economic Monitor," PlanEcon Report, vol. 9, nos. 19, 20, 21. June 10.
- Popiel, Walter 1993a. "Ukraine in Numbers. A Statistical Summary of Main Economic Indicators in Ukraine. Special: Year-End 1992 Edition." Cabinet of Ministers, Monthly Economic Bulletin, vol. 4, February.
- Popiel, Walter 1993b. "Ukraine in Numbers." Reprinted in The Ukrainian Legal and Economic Bulletin, edited by Alex Frishberg and David Snelbecker, vol.1, no. 5, Kiev, Ukraine (Available from the Project on Economic Reform in Ukraine, Kennedy School of Government, Harvard University.) Contact Ms. Alexandra Isaievych, (617) 496-8816.
- Popiel, Walter 1993c. "Ukraine in Numbers." Reprinted in The Ukrainian Legal and Economic Bulletin, edited by Alex Frishberg and David Snelbecker, vol.1, no. 6, Kiev, Ukraine.
- Transition: The Newsletter about Reforming Economies. Transition and Macro-Adjustment Division, Policy Research Department, The World Bank, vol. 4, no. 5, June 1993.
- U.S. Government 1992. Ukraine: An Economic Profile, 1992, U.S. Government Publications Office.